Calcium and Endothelium Dependent Responses of Aortic Vascular Smooth Muscles in Pregnant Rats

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ABSTRACT

This study compared the calcium dependent responses of aortic ring vascular smooth muscle and the effect of de-endothelization in pregnant and nonpregnant rats. Twenty adult female Sprague Dawley rats of comparable weight (200±40g) were maintained under standard laboratory conditions in two groups (n=10) consisting of a control and test (pregnant) rats fed on normal rat chow. At day 17 to 19 of pregnancy, rats were sacrificed and the aortic blood vessels vascular smooth muscle isolated (with intact and removed endothelium) and processed in organ bath for EC_{50} and maximum contraction strengths in different concentration of Ca^{2+} physiological salt solution (PSS). The data were subjected to statistical analysis (t-test) at $p<0.05$. Pregnancy caused a significant increase in the aortic ring response to phenylephrine in both 1.6mM and 0.8mm Ca PSS with significant higher EC_{50} in decreased Ca PSS. Removal of endothelium caused a significant decrease ($p<0.05$) in EC_{50} in both control and pregnant rats and was significantly higher in the pregnant rats compared to the control in both the endothelium intact and removed groups. Vascular smooth muscle contraction decreased significantly with decreasing Ca concentration and decreased significantly in pregnant rats compared to control in both 1.6mM (1909.50 ± 82.30 vs 1139.00 ± 103.40) and 0.8mM Ca PPS (1300.85 ± 57.36 vs 880.00 ± 25.95). However, removal of endothelium cause a non-significant increased ($p>0.05$) in the control (1909.5±82.3 vs 2012±63.4) but significant in the pregnant rats (1139.0±103.4 vs 2181±145.4). The findings therefore showed that Ca^{2+} concentration of PSS stimulates aortic vascular ring contractile responses to phenlyphrine. While pregnancy may attenuate this contractile response, the absent of endothelium may modulate it in pregnant rat but not in nonpregnant rat.

Keywords: Calcium, Pregnancy, Smooth muscle contraction, endothelium.

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1. INTRODUCTION

Normal pregnancy comes with extensive changes on the cardiovascular system that include a rise in blood volume and cardiac output, a fall in blood pressure and peripheral resistance as well as generalized vasodilatation to accommodate the pregnancy demands[1, 2]. These changes are likely to be induced by the hormonal influences[3]. There have been many reports on altered endothelial functionin normal pregnancy[4]. It is however unclear whether the altered levels of tone that are observed in normal pregnancy are due to changes in external influences on vascular smooth muscle, or intrinsic alterations in vascular smooth muscle cell signaling[1].

There is evidence of reduced contractility to stretch stimulus during pregnancy[5, 6], which is known to activate the voltage operated calcium channels[7] and increased potassium induced relaxation of isolated vascular smooth muscle[5, 6]. These changes could be due to the direct effect of pregnancy hormones on the vascular smooth muscle or indirect effect through blood borne agents[8, 9]. It is generally accepted that a rise in intracellular free calcium concentration mediates the contractile responses of all muscle types[10, 11]. The activating calcium necessary for the contraction is chiefly of dual origin: release from intracellular stores or influx from the extracellular medium[12]. Intracellular free calcium is the final trigger in the regulation of vascular smooth muscle contractility. The

sources of the activator calcium are influx from the extracellular medium through either the receptor operated calcium channel (ROC), activated by agonist such as phenylephrine, or voltage operated calcium channels (VOC), activated by membrane depolarization and release of calcium ions from the internal stores which accounts for phasic contractions[13], while tonic contractors are supported by Ca^{2+} ion influx from the extracellular medium.

The inner lining of blood vessel known as endothelium; is known to modulate autonomic and hormonal effects on the contractility of blood vessels. Furchgott and Zawadzki[14] demonstrated that intact endothelium was necessary to achieved vascular relaxation in response to acetylcholine. Preparations of rabbit thoracic aorta failed to elicit relaxation in response to acetylcholine following mechanical damage (rubbing the inner lining) of the endotheliumin several studies [15, 16]. Thus, both calcium and the endothelia state may be necessary for vascular smooth muscle contractility.This study compares the responses of aortic vascular smooth muscles to variable concentrations of calcium ion solutions as well as the impact of endothelia damage in pregnant and non-pregnant rats.

2. MATERIALS AND METHODS

2.1. Study Design: Twenty adult Sprague Dawley rats (weighing 250-300g) were used for this study. They were kept in well ventilated building and grouped according to the type of experiments.

Group 1 (sub groups a and b): This group consisted of 10 non pregnant adult virgin rats fed normal rat chow.

Group 2 (sub groups a and b): This group consisted of 10 pregnant rats fed with normal rat chow.

Sub groups a and b are groups of aortic rings in which the endothelium was intact and removed respectively.

The rats were fed, ad libitum, on the respective diets and had free access to clean drinking water. In the experiment which involved the use of pregnant rats, pregnancy was achieved by confining a male and a femalerat together for three consecutive days for the purpose of mating after which the male rat was separated from the female rat. The female rats were monitored for weight change, and palpated for presence of fetus/es [17, 18], which were suggestive of pregnancy.

2.2. Tissue Preparation: The rats were sacrificed by stunning and decapitation. The tissues were prepared using standard methods for the study of isolated blood vessels [19]. The descending aorta was quickly removed, put in a petri dish containing physiological salt solution (PSS) and cleared of adhering connective tissues. The aorta was then cut into 2mm ring segments and each segment was suspended between two L-shaped holders. The lower holder was fixed to the base of 20ml organ baths containing physiological salt solution (PSS) while the upper holder was connected to transducer coupled to Ugo Bassile recorder. The aortic rings were studied under standard organ bath condition of temperature at 37^0C, and pH 7.4 and the PSS was bubbled with 95% O_2, and 5% CO_2 gas mixture. The presence of functional endothelium was ascertained by the observation of at least 42% relaxation to 10^{-7}M acetylcholine. A passive resting tension of 1gm which was the tension at which maximum responses to 10^{-5}M phenylephrine is obtained was applied. An equilibration period of 90 minutes was allowed before the commencement of the measurements. During the equilibration period, the aortic rings were rinsed at about 30minutes interval. During this time the rings were stimulated with 10^{-7}M phenylephrine contraction response of rat aorta to 10^{-5} M Phenylephrine, following different levels of stretch.

2.3. Preparation of solution: Normal physiological salt solution: The composition of normal physiological salt solution was

(mM): NaCl, 119.0; KCl, 4.7 KH_2 PO_4, 1.2; $NaHCo_3$, 24.9. $CaCl_2$, 1.6; glucose, 11.5; Ca Na_2 EGTA 0.03.

Composition of 0.8ml physiological salt solution: It has the same composition as the normal physiological salt solution except that it contained 0.8mM $CaCl_2$ instead of 1.6mM.

Drugs**:** The drugs used in this study were phenylephrine hydrochloride (Sigma chemical Co. St Louis, M.O. USA); Phentolamine Mesylate (Rigitine Ciba); All drugs were prepared fresh using distilled water on each day of the experiment and kept-on Ice. Concentrations were expressed as final molar concentration (Mol/L) in the organ bath.

2.4. Concentration Response Test to Phenylephrine: The aortic rings from pregnant and non-pregnant rats were suspended in either normal PSS or 0.8mM $CaCl_2$ PSS, and exposed to increased concentration of phenylephrine (10^{-9}-10^{-4}M). This was done by cumulatively increasing the concentrations of phenylephrine in the bath. A subsequent application of phenylephrine was done when the effect of the earlier application had become stable. The contractile responses to phenylephrine of rings from pregnant and non-pregnant rats in the respective physiological salt solutions were compared. The protocol was also applied on rings of blood vessel from pregnant and non-pregnant rats from which the endothelium had been removed.

2.5. Statistics: Statistical analyses were carried out using the statistical package for social sciences (SPSS) version 21. Values are expressed in mean± standard error of mean (SEM). The was data analyzed using ANOVA to compare the differences between the groups. P-values less than 0.05 were considered significant.

3. RESULTS

Figure 1 compares the EC_{50} of the control and pregnant rats in 1.6 and 0.8mM Ca PSS. Decreased Ca concentration of PSS caused a significant increased ($p<0.05$) in EC_{50} in both control ($0.66 \pm 0.80 \times 10^{-7}$ vs $1.90 \pm 0.8 \times 10^{-7}$; p=0.0086; t=3.457; df=8) and pregnant rats ($2.47 \pm 0.31 \times 10^{-7}$ vs $4.10 \pm 0.31 \times 10^{-7}$; p=0.0001; t=8.173; df=8). The EC_{50} was significantly higher in the pregnant rats compared to the control rats in both the 1.6mM Ca PSS ($0.66 \pm 0.80 \times 10^{-7}$ vs $2.47 \pm 0.31 \times 10^{-7}$; p=0.0001; t=13.06; df=8) and 0.8mm Ca PSS ($1.90 \pm 0.8 \times 10^{-7}$vs $4.10 \pm 0.31 \times 10^{-7}$; p=0.0005; t=5.685; df=8).

Figure 2 compares the vascular smooth muscle contraction of the control and pregnant rats in 1.6 and 0.8mm Ca PSS. Vascular smooth muscle contraction decreased significantly with decreasing Ca concentration in PSS in the control (1909.50 ± 82.30 vs 1300.85 ± 57.36; p=0.0001; t = 13.57; df = 8) and pregnant rats (1139.00 ± 103.40 vs 880.00 ± 25.95; p = 0.0006; t = 5.433; df = 8) and contraction decreased significantly in pregnant rats compared to control in both 1.6mM (1909.50 ± 82.30 vs 1139.00 ± 103.40; p = 0.0001; t = 13.04; df = 8) and 0.8mM Ca PPS (1300.85 ± 57.36 vs 880.00 ± 25.95; p = 0.0001; t = 14.95; df = 8).

Figure 3 compares the EC_{50} of the control and pregnant rats with intact and removed endothelium in 1.6mM Ca PSS. Removal of endothelium caused a significant decreased ($p<0.05$) in EC_{50} in both control ($0.66 \pm 0.80 \times 10^{-7}$ vs $0.24 \pm 0.51 \times 10^{-7}$; p=0.0001; t=9.884; df = 8) and pregnant rats ($2.47 \pm 0.31 \times 10^{-7}$ vs $0.49 \pm 0.09 \times 10^{-7}$; p=0.0001; t=14.14; df=8). The EC_{50} was significantly higher in the pregnant rats compared to the control rats in both the endothelium intact and endothelium removed groups ($0.24 \pm 0.51 \times 10^{-7}$ vs $0.49 \pm 0.09 \times 10^{-7}$; p=0.0006; t=5.430; df=8).

Figure 4 compares the vascular smooth muscle contraction of the control and pregnant rats with intact or removed endothelium. Vascular smooth muscle contraction increased non-significantly ($p>0.05$) in the control (1909.5±82.3 vs 2012±63.4; p = 0.0584; t =

2.206; df = 8) but significantly ($p<0.05$) in the pregnant rats (1139.0±103.4 vs 2181±145.4; p = 0.0001; t = 13.06; df = 8) when the endothelium is removed. Comparatively, when the endothelium is removed, aortic vascular ring contraction significantly increased in the pregnant rat compared to the control (2012±63.4 vs 2181±145.4; p = 0.0444; t = 2.382; df = 8).

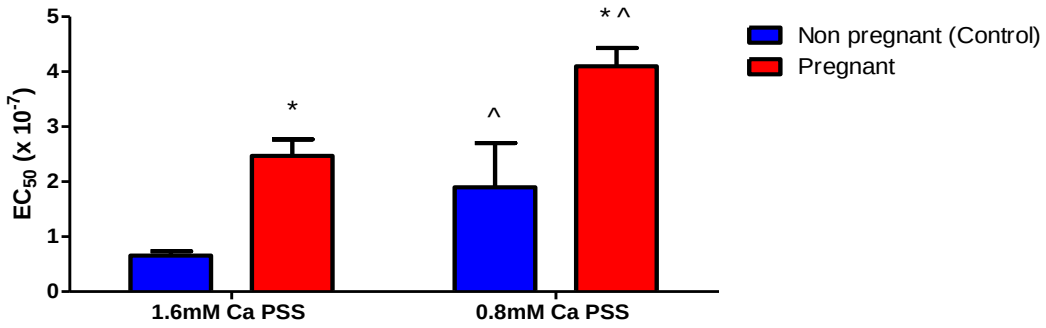


Figure 1. Aortic ring vascular smooth muscle response to phenylephrine in pregnant and non-pregnant rats in different Ca^{2+} concentration of PSS.

N=10 (n=5); Mean ± SEM; *significant at $p<0.05$ compared with control, ^significant at $p<0.05$ compared with 1.6mM Ca PSS.

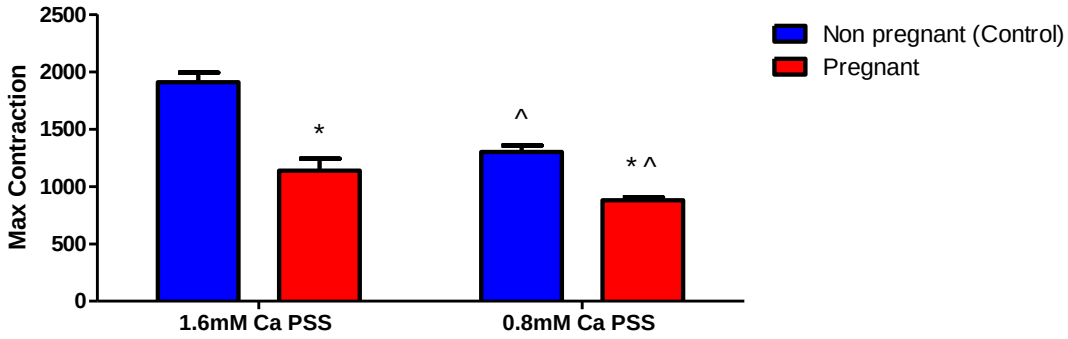


Figure 2. Vascular smooth muscle contraction of aortic ring to phenylephrine in pregnant and non-pregnant rats in different Ca^{2+} concentration of PSS.

N=10 (n=5); Mean ± SEM; *significant at $p<0.05$ compared with control, ^significant at $p<0.05$ compared with 1.6mM Ca PSS.

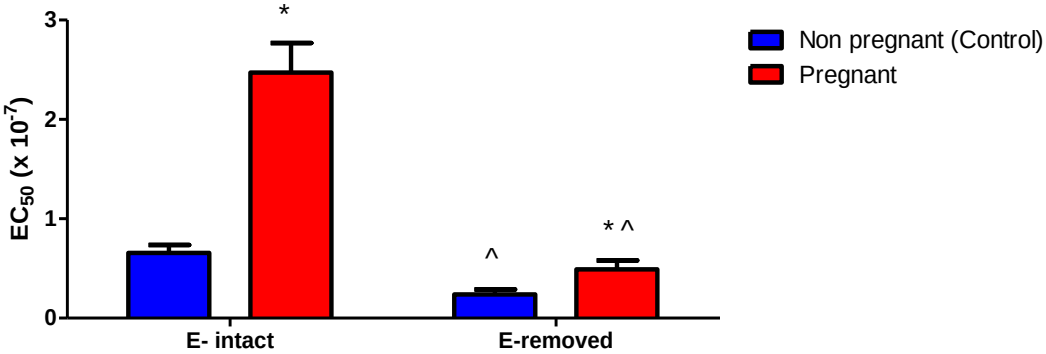


Figure 3. Aortic ring vascular smooth muscle response to phenylephrine in pregnant and non-pregnant rats with intact and removed endothelium.

N=10 (n=5); Mean ± SEM; *significant at $p<0.05$ compared with control, ^significant at $p<0.05$ compared with 1.6mM Ca PSS.

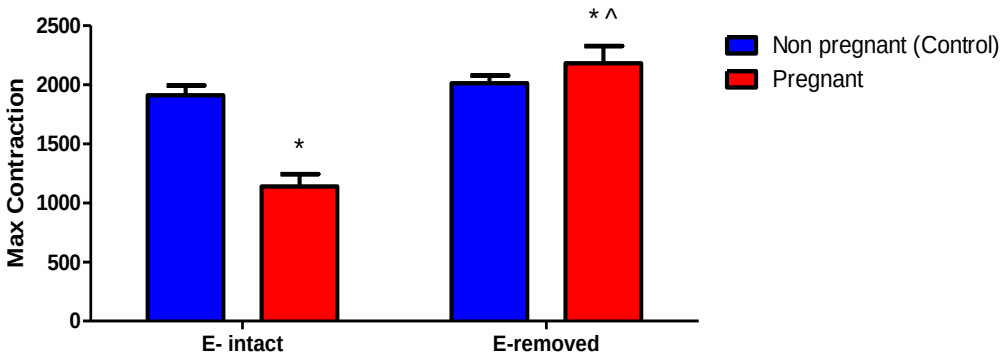


Figure 4. Vascular smooth muscle contraction of aortic ring to phenylephrine in pregnant and non-pregnant rats with intact and removed endothelium in normal PSS.

N=10 (n=5); Mean ± SEM; *significant at p<0.05 compared with control, ^significant at p<0.05 compared with 1.6mM Ca PSS.

4. DISCUSSION

In the present study, aortic ring smooth muscle contraction to phenylephrine was observed to increase with increased Ca^{2+} concentration in both the control and pregnant rat by 46.85% and 29.43% respectively while pregnancy cause a decreased in contraction by 67.65% and 47.82% in the 1.6mM and 0.8mM Ca PSS. The observations in the present study will give credence to the notion that a decrease in the concentration of Ca^{2+} PSS from 1.6mM to 0.8mM caused greater decrease in the contractions of aortic rings from non-pregnant and pregnant rats with greater effect in the non-pregnant rats. This will suggest a greater dependence of the smooth muscle cells contraction of the blood vessel (aortic ring) on extracellular Ca^{2+} ions. Transport systems present in cell membrane control the movement of sodium, potassium, calcium and other ions in order to maintain the marked difference in concentration of these ions on the outside and inside of cells, which in turn, provide the electrochemical gradient needed for various cell functions. It is well reported that the reactivity of blood vessel to most vasoactive agents in pregnancy is decreased [20-22]. The possibility that pregnancy produces its effect on vascular responses through inhibition of Ca^{2+} influx into vascular smooth muscle cells is not conclusively investigated and may not be overruled. On the other hand it is suggested that pregnancy may attenuate the contraction of blood vessel by decreasing calcium influx through the voltage operated calcium channels (VOCs) and the receptor operated calcium channel (ROCs) of vascular smooth muscle cells[23].

Normal pregnancy is accompanied with increased production of several potent vasodilators such as prostacyclin (PGI_2)[24] and nitric oxide[25, 26]which can have effect on how the vascular system function in this state. It has long been demonstrated that endothelium dysfunction alters the vascular relaxation response to acetylcholine considering the findings that rabbit thoracic aorta failed to elicit relaxation in response to acetylcholine following mechanical damage of the endothelium[14]. In the present study, endothelium damage caused increased contraction of aortic ring to normal PSS in both the non-pregnant and pregnant rats with significant different in the pregnant rats. This finding indicates that damage endothelium during pregnancy may cause increased contraction of vascular smooth muscle and in turn alter vascular tone homeostasis. This increased contraction may be due to the increased production of intrinsic and extrinsic vascular factors during pregnancy which may influence vascular smooth muscle.In support of the contention that the vascular endothelium is involved in the effect of

pregnancy on vascular responses to vasoactive agent[24, 27] was the observation that the attenuation of contractile responses to phenlyphrine in pregnancy was abolished by endothelium removal. Though the precise mechanism for this effect of pregnancy is not clear, the alteration of vascular endothelium function in pregnancy has been suggested and demonstrated[21, 25].

5. CONCLUSION

Extracellular Ca^{2+} concentration and pregnancy alter the contractile response of aortic ring vascular smooth muscle with de-endothelium in pregnancy modulating the contractile responses. Contractile responses of vascular smooth muscles, like in any other muscles are dependent on availability of Ca^{2+}. The dependence on extracellular Ca^{2+} is greater in non-pregnant rats than pregnant rats. Vascular responses are diminished in pregnancy and the mechanism by which this takes place appears to be endothelium dependent, as well as the vascular smooth muscle handling of extracellular calcium especially during pregnancy.

REFERENCES

1. VanWijk MJ., Boer K., van der Meulen ET.,Bleker OP., Spaan JAE. and VanBavel E. Resistance artery smooth muscle function in pregnancy and preeclampsia. *Am. J. Obstet. Gynecol.* 2002;186:148-54.

2. Robson SC.,Hunter S.,Boys RJ., DunlopW.Serial study of factors influencing changes in cardiac output during human pregnancy.*Am J. Physiol.*1989; 256 H1060-111065.

3. Goldberg C.A., Schrier RW. Hypertension in pregnancy.*Semin Nephroll*. 1991; 11:576-593.

4. Poston L. Maternal vascular function in pregnancy. *J Hum Hypertens*. 1996;10:391-4.

5. Mcaughlin MK.,KeveTM.Pregnancy induced changed in resistance blood vessels.*American Journal of Obstetrics and Gynecology*1986; 155, 1296- 1299.

6. Ebeigbe AB.,Ezimokhai M. Vascular smooth muscle responses in pregnancy – induced hypertension. *Trends in Pharmacological Sciences*1988; 9, 455-457.

7. William A.C. Voltage-Gated Calcium Channels. *Old Spring HarbPerspect Biol.* 2011; 3(8): a003947.

8. Enkhmaa D., Wall D., MehtaPK., StuartJJ., Rich-EdwardsJW, Bairey MerzCN., ShufeltC .Preeclampsia and Vascular Function: A Window to Future Cardiovascular Disease Risk.*J Womens Health (Larchmt) 2016;*25(3): 284–291.

9. TuienkoT., SchneiderJ.,Floro C.,Sicilia M.The in vitro effect on arterial wall function of serum from patients with pregnancy induced hypertension. *America Journal of Obstericsand Gynecology*1987;156, 17 - 23.

10. Ivana YK., Barbara EE. Signaling in muscle contraction.*Cold Spring HarbPerspect Biol.* 2015; 7(2): 06023.

11. Bohr DF. Vascular smooth muscle update.*Circulate Res.*1973; 32, 665-672.

12. Godfraid T.,Kaba A. *The* role of calcium in *the* action of drugs on vascular smooth muscle. *Arch. Int. Pharmacodyn*1972; 196: 35-49.

13. Fabiato OA,Fabiato F. Calcium release from the sarcoplasmic reticulum. *Circulat. Res.*1977; 40:119-129.

14. Furchgott RF., Zawadzki JU. The obligatory role of endotheliab cells in the relaxation of arterial smooth muscle by acetylcholine.*Nature*1980; 288:373-376.

15. De Mey JG., Vanhoutte PM. (1981). Role of the intima in cholinergic and purinergic relaxation of isolated canine femoralarteries J. Physiol. 1981; 316:437-455.

16. Ebeigbe AB.,Aloamaka CP. Role of endothelium in magnesium – induced relaxation of rat aorta. *Res. Exp. Med.*1987; 187:25-31.

17. Agematsu Y., Ikadai H.,Amao H. Early detection of pregnancy of the rat.*JikkenDobutsu*.1983; 32(4):209-12.

18.Smarr BL., Zucker I., Kriegsfeld LJ. Detection of successful and unsuccessful pregnancies in mice within hours of pairing through frequency analysis of high temporal resolution core body temperature data. PLoS ONE 2016; 11(7): e0160127.

19. Touyz RM. Reactive oxygen species as mediators of calcium signalling by angiotensin II: implications in vascular physiology and pathophysiology. *Antioxid Redox Signal* 2005; 7:1302–1308.

20. Funchs KL., Grindlay–Moore L, Rounds S. Pulmonary vascular reactivity is blunted in pregnant rats. *Eur. J. Pharmacol*1982; 25: 267-269.

21. Massiocotte G., Louis J.,Parent A., SchiffrinEI. Decreased in vitro responses to vasoconstrictors during pregnancy in normotensive and spontaneously hypertensive rats Can. *J. Physiol. Pharmacol.*1987; 65: 2466 -2471.

22. Magness RR., Roseufeld CR. Mechanism of altered pressor responses to alpha agonists in ovine pregnancy. *Am J. ObstetGynecol*1988; 159: 252 -- 261.

23. Ezimokhai M, Aloamaka CP.,Chjerian T., Morrison. The role of extracellular calcium in pregnancy – induced attenuation of phenylephrine contraction in rat aorta with functional endothelium.*J. Comp. Physiol. B.*1994; 164:81-87.

24. Goodman RP., Kulam AP., Brash AR., Branch RA. Prostacycline production during normal pregnancy and pregnancy complicated by hypertension. *AMJ Obstet. Gyneiol.*1982; 142:817-822.

25. Ahokas RA.,Mereer BM., Sibai BM. Enhanced endothelium-derived relaxing factor activity in pregnant, spontaneously hypertensive rats. *American Journal of Obstertricand Gynecology*1991; 165, 801-807.

26. Weiner CM., Zhu LK.,Ghodsi CD. In vitro release of endothelium – derived relaxing factor by acetylholine is increased during the guinea pig pregnancy Am J ObstetGyneacol1989; 161:1599-1605.

27. Aloamaka C.P., Ezimokhai M, Cheriasn T., Morrison JM. Mechanism of pregnancy-induced attenuation of contraction to phenylephrine in rat aorta. *Exp. Physical*1993; 78:403-410.